



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

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Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

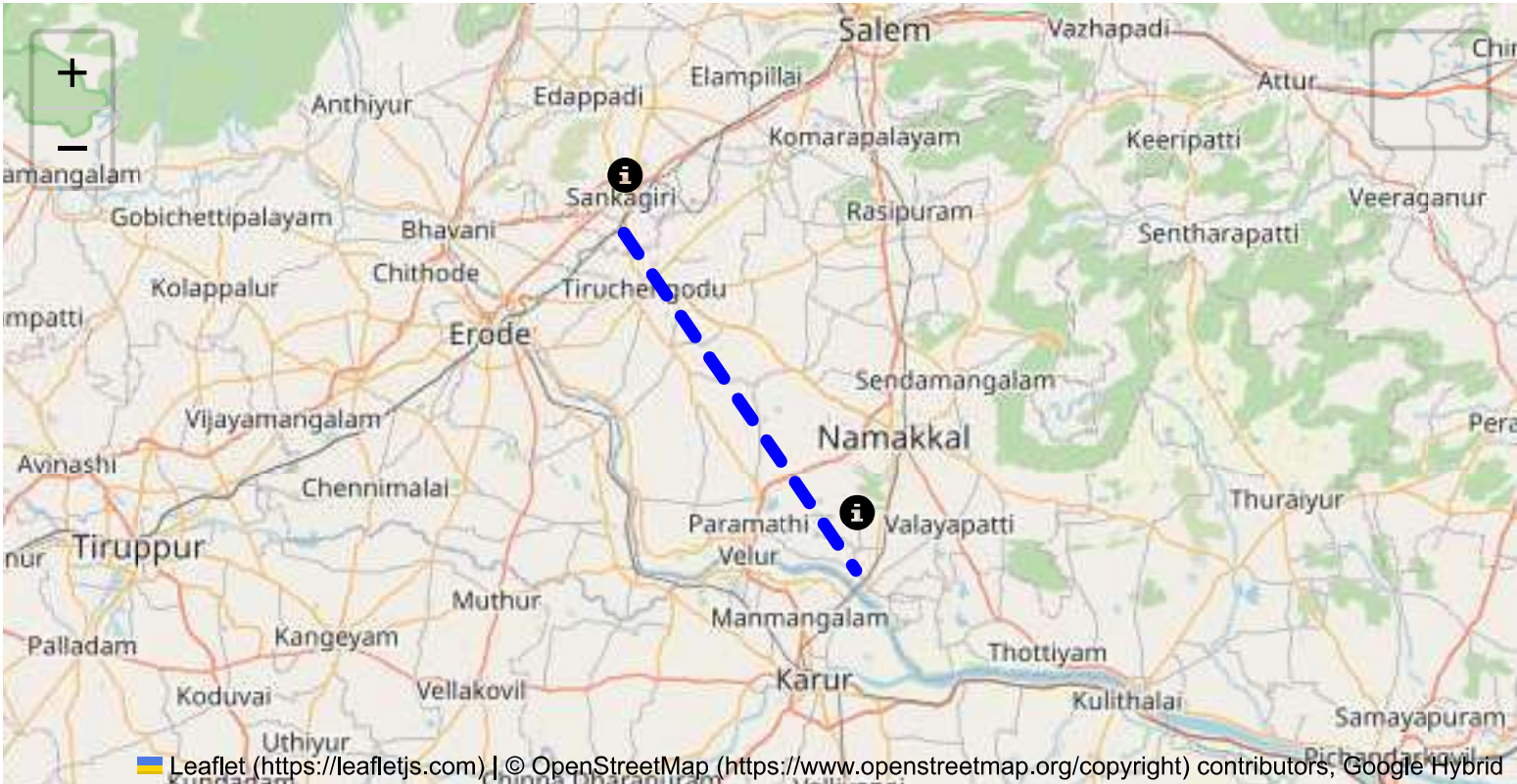
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

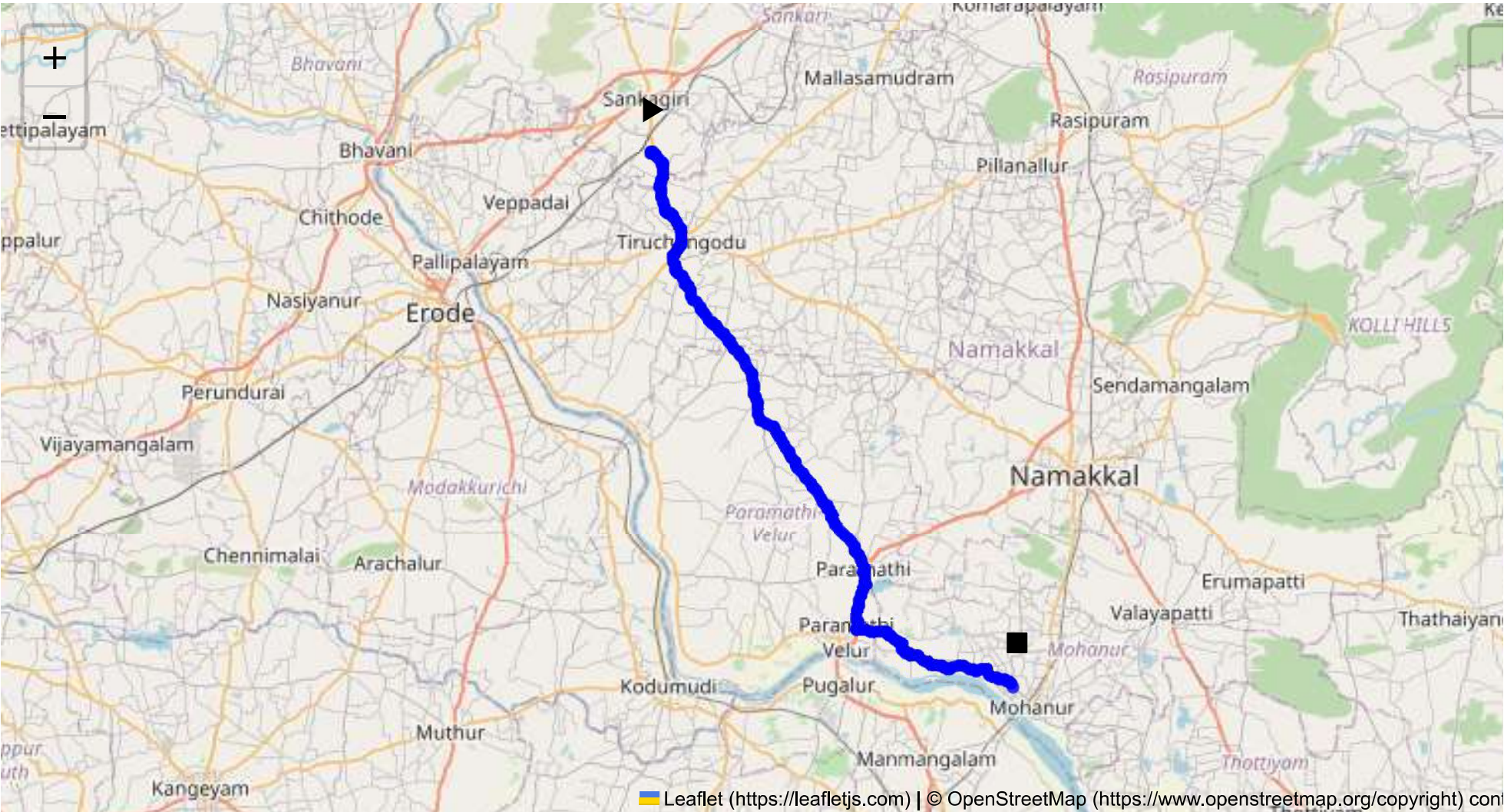
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.28 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.075236, 78.125893)



Welcome to the Journey Risk Management Study

Route Safety Analysis: CVQF+23W, Sangagiri to 34GG+39P, Velur Rd, Tamil Nadu, India

1. Overview of the Route Map:

- The route from Sangagiri to Velur Rd covers approximately 56.28 kilometers, primarily utilizing NH 44 and SH 79. This route passes through semi-urban and rural areas, with diverse landscapes including agricultural fields and small towns.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The region experiences a tropical climate, with hot summers, moderate monsoons, and mild winters.
- During the monsoon season (June to September), heavy rains can lead to flooding and reduced visibility. Therefore, caution is needed to avoid hydroplaning and manage slippery roads.
- Rising temperatures in summer may induce vehicle overheating, requiring regular maintenance checks.

3. Analysis of Traffic Patterns:

- Peak traffic hours typically occur during morning (8:00 - 10:00 AM) and evening (5:00 - 7:00 PM), with congestion expected near major junctions and urban centers.
- Traffic is usually dense in areas such as the entry and exit points of towns along NH 44. Additional caution is advised during market days when local traffic increases.

4. Assessment of Road Quality and Infrastructure:

- NH 44 is generally in good condition, with a well-maintained surface suitable for heavy vehicles. However, SH 79 may have uneven patches and narrower lanes.
- Drivers should be alert for occasional potholes, and construction zones, especially during monsoon repair works.

5. Suggestions for Alternative Routes for Emergencies:

- In case of route blockage or severe congestion, consider taking local roads parallel to NH 44 and SH 79, ensuring to rejoin the main route where traffic normalizes.
- GPS navigation should be used to identify the least efficient detour paths due to varying road quality.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transport of hazardous materials is regulated under the Motor Vehicles Act, requiring specific permits and adherence to safety protocols like displaying proper placards and maintaining safe speeds.
- Heavy fines or vehicle impoundment can occur if these regulations are not followed.

7. Overview of Historical Incidents:

- Historical data indicates occasional incidents involving heavy vehicles, primarily due to brake failures or driver fatigue. Hence, ensuring vehicle maintenance and driver alertness is crucial.

8. Environmental Considerations and Sensitive Areas:

- The route traverses agricultural regions and proximity to water bodies, necessitating careful handling of hazardous materials to prevent environmental contamination.
- Local communities are positioned along the route, emphasizing the need for sound traffic management and accident readiness.

9. Analysis of Communication Coverage:

- Mobile network coverage is typically strong along NH 44 but may experience brief interruptions in more rural stretches or underpasses.

- Ensure all communication devices are charged, and consider using satellite phones for reliable connections.

10. Estimated Emergency Response Times:

- Emergency response can take 20-30 minutes in urban areas due to the proximity of services. In rural stretches, response times may extend up to an hour.
- Users should have contact details of local emergency services readily available.

11. Overall Summary of Risk Assessment:

- The route is generally suitable for transporting hazardous materials with proper safety measures. Key risks include local traffic congestion, variable road quality, and weather conditions, especially during monsoons.
- Preparation, continuous risk assessment, and route monitoring are essential for mitigating potential incidents. Visibility improvements and signage reinforcement can assist in safer navigation.

In conclusion, while some challenges exist, they can be managed with proactive planning, adherence to regulations, and real-time responsiveness to changing conditions along the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16764, 78.01565	30 KM/Hr
7	Turn	High	11.15573, 78.02176	15 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	High	11.11887, 78.01758	15 KM/Hr
11	Turn	High	11.11883, 78.01771	15 KM/Hr
12	Blind Spot	Blind Spot	11.11397, 78.01722	10 KM/Hr
13	Turn	Medium	11.10418, 78.04796	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	SCSM Polytechnic college	11.0786881, 78.1277794	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.11887, 78.01758



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.11883, 78.01771



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.11397, 78.01722



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10418, 78.04796

